

Dane Z. Dalton

Location: Huntsville, AL

Contact: (224) 829-8505 | danezdalton1@gmail.com

Professional Objective

With a background spanning digital circuits, CAD, controller programming, and software development, I've cultivated a deep passion for technology. I relish the challenges software brings, finding immense satisfaction in reaching my own creative solutions that simplify work-flow for colleagues and clients. Equipped with a diverse skill set, what I have to offer is potential. As a solo frontend developer on my current project, I've practiced the ability to learn whatever is needed to produce new features for my client.

Education

Iowa State University, Ames, IA | 2021 | Bachelor of Science, Environmental Science | Minor in Biology

Work Experience

Adventures With Mr. Math, Remote - *Software Developer, Director of Teaching Assistants (July 2022 - Present)*

- Build a multi-faceted E-Learning Portal for over one thousand students, parents, TA's, and teachers of Mr. Math. Full life cycle development with ReactJS and TailwindCSS as the sole frontend developer, Dockerified for production.
- Functionality: sign in, sign up, auth, class scheduling/rescheduling, customized interactive worksheets, and a mailbox system.
- Designed a child-friendly UI/UX for a primarily Elementary/Middle Schooler user-base.
- Engineered a client-side caching system for the call-heavy worksheets. This noticeably improves the fluidity of working between multiple worksheet problems at the same time, and reduces stress on the server.
- Organize and streamline the dev experience with 3rd party tools including: Prettier, MaterialUI, date-fns, react-icons, and Axios.
- Create and Lead a QA Process. Collaboration with a Lead QA Engineer helped me mold this process to our needs.
- Scheduling, management, administration, organization, and interviewing skill growth managing a 60+ TA team.
- Create a C# program to calculate the TA hours at the end of each month using the Google Sheets API.

Citizens For Conservation, Barrington, IL - *Restoration Intern Crew Lead (Summer 2020)*

- Lead a restoration crew to carry out prairie, wetland, and savannah restoration management plans.
- Lead volunteer work days.
- Experience extracting and inputting mapping data with ArcGIS.

Projects & Extracurriculars

My Portfolio, <https://www.danezdalton.com/> : A small portfolio to showcase my passion for creativity, and some of the projects I've worked on. While I built this website to practice TypeScript, I've enjoyed improving upon some code and design choices I made for the Mr. Math Portal. A more exciting experience on desktop, but mobile is fine too.

Custom LED Light String: Decorative LED strip wired to an Arduino, using Arduino code (pseudo C++) and RGB values to create different light patterns. A semester-long project class, entirely self-directed and self-taught.

Etch-A-Sketch: A web app written in JS that provides an interactive color palette with an opacity setting that allows for modification of a sketch grid on a 'mouseover' event. The user/child friendly UI/UX led to the opportunity to create the Mr. Math portal frontend.

Knight Travails: An application written in Ruby using Graph Theory to find the fastest path for a knight in a chess game to move from one square to another. Building this graph breadth first, the first occurrence of a finished game is found and the parent nodes are traced to output the quickest path from square A to square B.

Rock Climbing: At my local climbing gym I routeset one day a week. Creating challenges that other climbers get to explore and experience is incredibly rewarding. What I love about climbing is the opportunity to spend all day outdoors, set goals, studying every little pebble to get the most out of every hold, and perfecting my movement to top the climb exerting as little energy as possible.

Skills

Software Skills: ReactJS, TailwindCSS, JavaScript, TypeScript, HTML, CSS, UI/UX, Docker, Git, Vite, C#, SQL, SSMS, QA Lead, Ruby, Rails, Linux OS, REST API, Postgres, Bootstrap, C, C++, CAD, GIS, LaTeX, Salesforce Certified Administrator, Excel, Google Workspace, Godot, GDScript

Hardware Skills: Arduino, Logic Gates, Solder, Breadboard, PC Building, Multimeter, Oscilloscope, Isometric Sketching, Autoclave